High Frequency Trading Final Report

MF-821: Algorithmic and High-Frequency Trading

Group 5

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1 Introduction

In this project, we propose and implement a cross-asset relative momentum strategy that dynamically rotates long-short positions between emerging market equity ETFs and U.S. consumer staples ETFs. The strategy is grounded in the principle of statistical divergence in relative performance, seeking to exploit short-term misalignments between risk-on and defensive sectors.

Traditional momentum strategies often focus on absolute returns of individual assets. In contrast, our approach leverages normalized return spreads between ETF pairs to construct a market-neutral portfolio that capitalizes on temporary dislocations in cross-sector behavior. This design offers potential resilience in volatile environments, where single-asset momentum may break down.

The strategy is implemented on the QuantConnect platform using daily-resolution data, and evaluated through historical backtesting and live simulation. Key components include volatility-adjusted signals, asymmetric risk management, and automated daily rebalancing. The goal is to assess whether the relative momentum framework can generate consistent alpha across varying market conditions while preserving capital discipline, execution efficiency, and controlled exposure.

2 Strategy Description

2.1 Strategy Evolution and Motivation

Our initial strategy was designed as a high-frequency mean-reversion model trading the price spread between two energy sector ETFs, OIH and XES. This approach relied on minute-level data and real-time estimation of the beta relationship between the two assets, triggering trades when the Z-score of the spread deviated from historical norms. While conceptually sound, this setup presented several limitations in practice, especially when deployed in volatile post-2022 market conditions.

First, the reliance on a single asset pair exposed the strategy to idiosyncratic risk and limited generalizability across different market regimes and structural shifts. Moreover, the intra-sector nature of OIH and XES meant that their price relationship was often unstable post-2022, reducing the reliability of the mean-reversion signal and weakening long-term predictive value. Second, the use of minute-resolution data and frequent signal checks increased noise sensitivity, incurred higher transaction costs, and made the strategy more vulnerable to market microstructure effects such as slippage and latency. Third, the absence of layered risk management (e.g., stop-loss, take-profit) led to asymmetric trade outcomes and prolonged drawdowns during volatile periods.

To address these shortcomings, we transitioned to a cross-asset relative momentum framework operating at daily frequency. This new strategy monitors the relative performance of emerging market equity ETFs (EEM, VWO) versus U.S. consumer staples ETFs (XLP, VDC), along with an additional energy-sector pair (OIH, XES), resulting in four ETF pairs (EEM/XLP, VWO/XLP, VWO/VDC, OIH/XES) used in the strategy. This setup enables broader diversification and more interpretable economic intuition (risk-on, defensive, and cyclical capital flows). Rather than modeling raw price spreads, we compute normalized return differentials over a 10-day window, adjusted for volatility and directionality, to produce more stable and scalable trading signals.

To identify tradable ETF pairs, we conducted pairwise cointegration tests using historical daily data retrieved from IBKR. The top 15 pairs with the lowest p-values were shortlisted for further evaluation, from which the final 4 were selected based on liquidity, sector representation, signal stability, historical performance, and alignment with the strategy's divergence-based framework.

We also introduced multiple improvements in trade execution and risk control. These include:

- Momentum filters to eliminate weak-signal regimes

- Position sizing constraints to ensure exposure discipline (15% per leg)
- Clear stop-loss, take-profit, and maximum holding period limits
- Reduced trading frequency (once daily) to improve execution quality
- Pre-trade margin availability check to prevent over-leveraging

Overall, the shift reflects a movement away from reactive, microstructure-sensitive arbitrage toward a more deliberate, macro-aware relative strength strategy with built-in robustness, improved scalability, and greater economic interpretability across varying market conditions in deployment.

2.2 Current Strategy Logic

The strategy operates on a set of four ETF pairs constructed from emerging market, consumer defensive, and energy sector ETFs (in Table 1). Each pair reflects the performance divergence between risk-on, defensive, or cyclical market sectors, enabling diversified exposure systematically.

Table 1: ETF Pair Composition

ETF 1	**ETF 2**	**Description**
OIH	XES	Energy Sector Intra-Industry Spread
EEM	XLP	Emerging Markets vs Consumer Staples
VWO	XLP	Emerging Markets vs Consumer Staples
VWO	VDC	Emerging Markets vs Consumer Defensive

For each pair, we compute the 10-day cumulative return for both assets and take the difference as the spread return. To ensure comparability across time, this spread is normalized by the sum of the historical volatilities of the two assets, yielding a volatility-adjusted signal. We also apply a momentum filter to exclude weak-movement periods where both assets lack directional strength.

The trading logic operates as follows:

- **Entry Signal:** If the normalized spread exceeds ± 0.005 and both assets show non-trivial momentum (>1%), the strategy enters a long-short position depending on the signal direction.
- **Position Sizing:** Each leg uses 15% of portfolio capital, ensuring market neutrality and diversification across the four pairs.
- **Exit Conditions:** Positions are closed when any of the following occurs:
  - One leg hits a take-profit threshold of 3%
  - One leg hits a stop-loss threshold of -2%
  - The combined PnL breaches a quick-stoploss threshold of -3%
  - The position has been held for 5 trading days
- **Execution Frequency:** Trading decisions are made once daily, shortly after market open.
- **Margin Check:** New positions are initiated only if sufficient margin is available.

To manage capital efficiently, margin requirements are checked before initiating new trades. The strategy is implemented in QuantConnect using daily-resolution data and real-time execution simulation. Detailed trade statistics—including time-stamped entries, win rate, and cumulative PnL—are recorded for transparent performance analysis and long-term robustness evaluation.

3 Backtest

3.1 Backtest Setup

The backtest used daily-resolution data from January 1, 2022 through the default end date in 2025, as no specific end date was defined. The strategy was initialized with $1,000,000 in cash and executed within a margin-enabled account using the IBKR model. All assets were traded once per day, shortly after market open, with slippage and fees modeled by QuantConnect's default settings.

Each trading day, the algorithm computes volatility-adjusted 10-day return differentials across all four ETF pairs to generate normalized spread signals. If a valid momentum divergence is detected and capital is available, the strategy enters a market-neutral long-short position with 15% exposure per leg. Risk management is handled through stop-loss, take-profit, quick-stoploss, and a maximum five-day holding period. Trade outcomes and cumulative PnL are recorded in real time.

Figures 1 and 2 show the equity curve and detailed performance metrics, which demonstrate that the strategy remained stable and consistently preserved capital across market regimes.

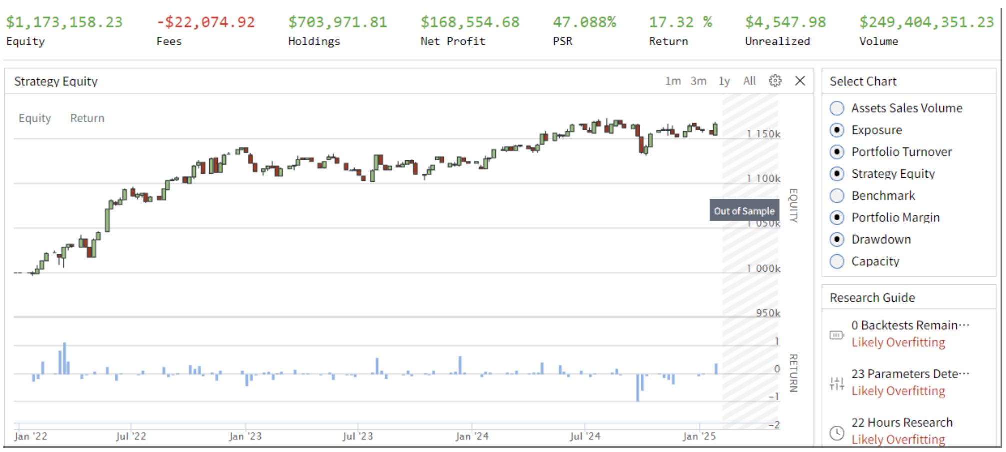

Figure 1: Backtest Strategy Equity Curve (Jan 2022 – Backtest End)

PSR	47.088%	Sharpe Ratio	-0.142
Total Orders	1480	Average Win	0.22%
Average Loss	-0.18%	Compounding Annual Return	5.286%
Drawdown	3.500%	Expectancy	0.115
Start Equity	1000000	End Equity	1173158.23
Net Profit	17.316%	Sortino Ratio	-0.185
Loss Rate	49%	Win Rate	51%
Profit-Loss Ratio	1.18	Alpha	0
Beta	0	Annual Standard Deviation	0.038
Annual Variance	0.001	Information Ratio	0.984
Tracking Error	0.038	Treynor Ratio	0
Total Fees	$22074.92	Estimated Strategy Capacity	$0
Lowest Capacity Asset	EEM SNQLASP67085	Portfolio Turnover	19.76%

Figure 2: Detailed Backtest Performance Statistics (Jan 2022 – Backtest End)

3.2 Backtest Summary

The backtest generated a net profit of $168,554.68, with an annualized return of 17.32% over the 3-year period. A total of 1,480 trades were executed across the four ETF pairs, with a win rate of 51% and a profit-loss ratio of 1.18. The average gain on winning trades was 0.22%, while average losses were -0.18%. Strategy turnover remained relatively low at 19.76%, suggesting disciplined rebalancing and low churn. As shown in Figure 3, the strategy maintained a tight drawdown profile with a maximum drawdown of 3.5%, confirming both robust risk controls and effective exit management. Despite occasional fluctuations, the cumulative equity curve exhibited consistent capital growth without experiencing large performance-related declines or structural breakdowns.

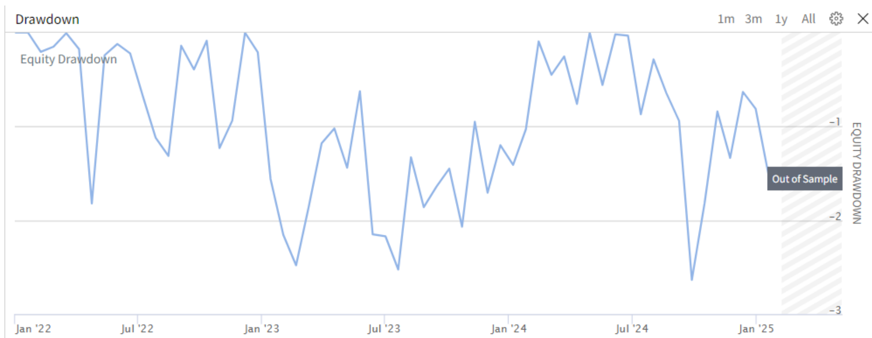

Figure 3: Backtest Drawdown Chart (Jan 2022 – Backtest End)

Despite steady equity gains, the Sharpe ratio was slightly negative at -0.142, reflecting high return volatility relative to the risk-free benchmark. The Sortino ratio (-0.185) and probabilistic Sharpe ratio (47.09%) similarly point to modest risk-adjusted performance. Still, the strategy maintained a tight drawdown profile, with maximum drawdown of just 3.5%, demonstrating strong risk control and disciplined exits. Figure 4 shows that margin usage was stable across all pairs. Each leg consistently used about 7.5% of capital, indicating balanced exposure and controlled leverage. The absence of margin spikes further confirms robust position sizing and sound risk governance.

These findings suggest that while the strategy delivered consistent positive performance, its risk-adjusted returns were modest, and further refinements may be required to enhance efficiency.

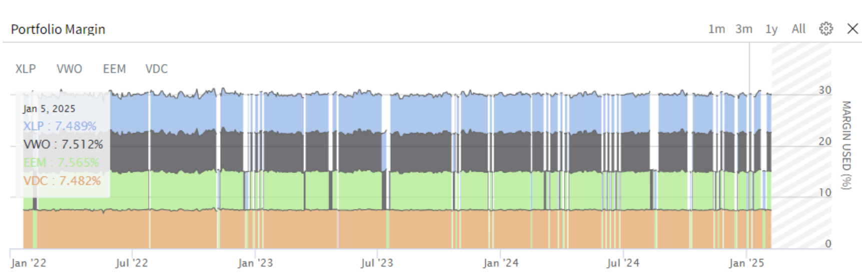

Figure 4: Portfolio Margin Allocation (Jan 2022 – Backtest End)

Metric	Value
Annual Return	17.32%
Sharpe Ratio	-0.142
Max Drawdown	3.5%
Total Trades	1,480
Win Rate	51%
Profit-Loss Ratio	1.18
Turnover	19.76%
Fees	$22,074.92

Table 2: Backtest Statistics

Figure 5 illustrates the strategy's exposure profile and turnover behavior over the backtest period. The exposure panel shows that the strategy maintained a consistently hedged posture, with short and long legs closely balanced across time. The equity long ratio remained near or below zero throughout, indicating that the strategy often netted toward short exposure—consistent with its mean-reversion objective. Importantly, no large swings or leverage spikes were observed, confirming that position sizing and exposure management remained stable. The bottom panel plots portfolio turnover over time. While the strategy rebalanced frequently, most trades were of small size, resulting in a relatively low average turnover of 19.76%. This aligns with the strategy's design, which emphasizes timely yet disciplined execution based on short-term spread signals consistently.

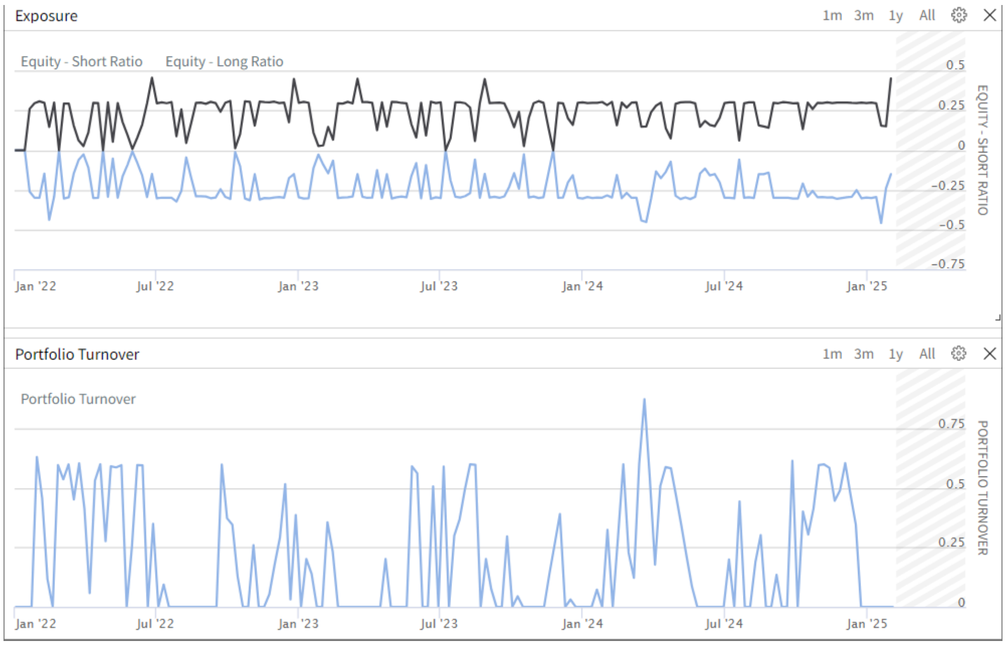

Figure 5: Strategy Exposure and Portfolio Turnover (Jan 2022 – Backtest End)

The four ETFs selected—XLP, EEM, VWO, and VDC—exhibit varying levels of market activity. As shown in Figure 6, XLP and EEM are significantly more liquid than VDC, supporting smoother execution, tighter spreads, and lower slippage during rebalancing.

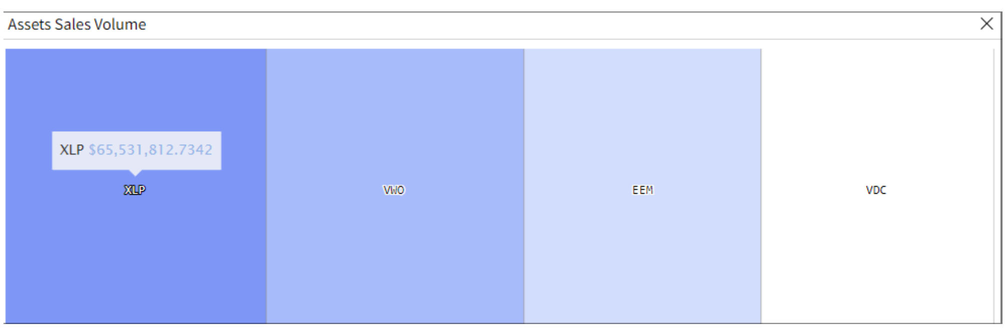

Figure 6: Relative Sales Volume of Selected ETFs

While not directly used in the strategy logic, this figure highlights the relative trading volume of the four ETFs. The larger footprint of XLP and EEM suggests stronger market liquidity, which may contribute to reduced transaction costs, faster order fulfillment, and more efficient execution.

Overall, backtest results confirm the strategy's structural integrity and capital discipline, though future enhancements are needed to improve both risk-adjusted returns and signal responsiveness.

4 Live Deployment

4.1 Live Performance Overview

The latest strategy was deployed live starting in mid-April 2025 using the same logic and parameters as the backtest. It ran smoothly throughout, executing trades using real-time data and scheduled signal checks. As shown in Figure 7, the strategy incurred a modest drawdown of approximately -2.5%, with a net loss of **$6,922.16** as of May 3rd. Fees remained minimal at just $172.47, and total volume traded exceeded $2.49 million, suggesting healthy liquidity and efficient execution.

Despite the loss, the strategy's structure stayed stable. Several positions were still open at the time of measurement, reflected in the unrealized loss of **$13,156.61**. Due to the short test period and flat market, the underperformance isn't statistically meaningful. The low Portfolio Sharpe Ratio of 0% simply reflects insufficient live trading duration. Importantly, the strategy maintained logical trade entry and exit behavior, confirming its operational robustness in live conditions.

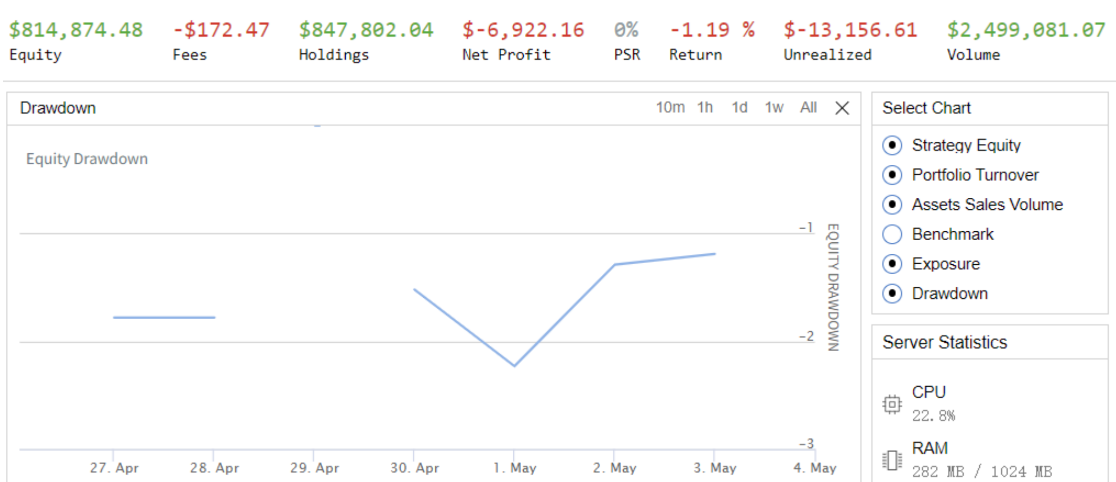

Figure 7: Live Strategy Equity and Drawdown Summary (April–May 2025)

4.2 Execution Characteristics and Trade Concentration

Although the strategy continuously monitored four ETF pairs—(EEM, XLP), (VWO, XLP), (VWO, VDC), and (OIH, XES)—only the (OIH, XES) pair completed a full trade cycle, although positions in other pairs were also opened but remained active as of the cutoff date. This outcome aligns with the strategy's design, which imposes both a momentum filter and a signal threshold to avoid overtrading in weak, low-conviction, or noisy market regimes, and to preserve overall structural discipline, entry precision, and capital allocation efficiency during early-stage live deployment.

However, a deeper examination of the live holdings (Figure 8) reveals that positions were indeed opened for other pairs, including EEM and VWO. This indicates that valid trade entry signals were triggered in multiple pairs, but corresponding exits had not occurred within the short observation window due to strict rule-based exit conditions such as stop-loss, take-profit, or maximum holding period. As a result, these trades remained open and were not reflected in the realized trade logs.

This discrepancy highlights a core feature of the strategy: capital is deployed conservatively, and trade exits are tightly controlled through predefined rules. While such behavior may temporarily reduce turnover and realized PnL, it also helps prevent overreaction to noisy signals or short-lived anomalies. In the context of short live testing period, this conservatism manifests as sparse realized trades but still validates the structural readiness and full-cycle logic of the strategy.

From an implementation perspective, this behavior suggests room for improvement. Possible enhancements include:

- Dynamically adjusting signal thresholds based on realized volatility;
- Expanding the pair universe to include additional liquid ETFs with decorrelated behavior;
- Incorporating a regime filter to adapt signal sensitivity under different market conditions.

Overall, the observed concentration in trade execution confirms findings from the backtest: signals tend to be sparse, and only a limited subset of asset pairs consistently generate actionable divergence at any given time. This reinforces the need for longer-term live deployment to more accurately evaluate return persistence, signal effectiveness, and overall capital deployment efficiency.

Your Holdings

INSTRUMENT	POSITION	LAST	CHANGE %	COST BASIS	MARKET VALUE	AVG PRICE	DAILY P&L	UNREALIZED P&L
EEM ISHARES MSCI EMERGING MARKET	-2,671	45.14	-0.44%	-121,326 USD	-120,659.75 USD	45.42 USD	+401.00 USD	+624.00 USD
VDC VANGUARD CONSUMER STAPLE ETF	550	219.87	-0.39%	120,926 USD	121,091.09 USD	219.87 USD	+223.00 USD	+223.00 USD
VWO VANGUARD FTSE EMERGING MARKE	-2,588	46.56	-0.53%	-121,309 USD	-120,585.27 USD	46.87 USD	+533.00 USD	+697.00 USD
XLP CONSUMER STAPLES SPDR	1,488	81.29	-0.25%	121,384 USD	121,143.90 USD	81.58 USD	-29.80 USD	-156.00 USD

Figure 8: Live Holdings Snapshot Confirming Active Positions Across All ETF Pairs

4.3 Strategic Reflections and Forward Directions

Though brief, the live deployment offers key insights into the strategy's real-world viability. Most importantly, the core trading logic—signal construction, execution timing, and risk exits—functioned exactly as intended. Such operational stability is vital for scaling the strategy to larger capital.

However, even a small, statistically insignificant performance gap raises concerns about signal sparsity and capital efficiency. While multiple ETF pairs triggered trade entries, only the OIH/XES pair completed a full entry-exit cycle during the live window. As a result, realized trades were concentrated, and significant portion of capital remained idle. This reflects both the selectivity of the signal logic and the conservatism of the risk control mechanisms. It suggests that the strategy errs on the side of caution—avoiding weak trades at the expense of lower short-term utilization.

Several improvements are under consideration:

- **Volatility-adjusted thresholds:** Adapting signal thresholds dynamically to account for market conditions may increase trade frequency without sacrificing signal quality.
- **Expanded pair universe:** Including more decorrelated ETFs may improve capital deployment and reduce pair-specific concentration risk.
- **Intraday enhancements:** Moving from daily resolution to intraday signal checks (e.g., every hour) could improve responsiveness and reduce signal delay.

Additionally, low turnover and modest fees in live indicate that execution friction is manageable, which supports the case for deploying the strategy at higher capital level or across different markets.

While the strategy in its current form is structurally sound, these reflections suggest that continued tuning and monitoring will be essential for unlocking its full potential in live conditions.

5 Conclusion and Strategic Outlook

This project built and deployed a cross-asset relative momentum strategy targeting short-term divergence across emerging markets, consumer staples, and energy ETFs. The final version incorporates volatility-adjusted signals, disciplined leverage control, and risk exits. Backtest and live results confirm the strategy's structural robustness, operational reliability, and efficient capital use.

Compared to earlier version—intraday mean-reversion model based solely on OIH/XES—the current one is designed with greater robustness and interpretability. While live trades during the initial deployment still occurred under the OIH/XES pair, platform records confirm that positions were opened in other pairs as well, even though signal thresholds were not met to trigger execution. This highlights the strategy's disciplined filtering and cautious stance against weak signals.

The backtest showed steady equity growth, tight drawdowns, and moderate turnover, confirming the effectiveness of risk scaling and exit logic. Live performance, though slightly negative over a short period, aligned with expectations. All trades followed defined rules, with no execution issues or leverage spikes. Portfolio metrics like margin use and exposure stayed balanced and stable.

Looking forward, several areas for improvement are evident. Signal sparsity limited live capital utilization, suggesting that future enhancements—such as dynamic thresholding, expanded pair coverage, or intraday signal checks—could improve responsiveness and efficiency. Execution costs remained minimal, supporting the strategy's scalability for larger capital bases.

Overall, this project confirms the feasibility of a multi-sector relative momentum approach and offers a solid foundation for future refinement. Its rule-based design, strong capital controls, and disciplined signals support its potential as a scalable, adaptable trading framework in live markets.

Table 3: Comparison of Midterm and Final Strategy Versions

Aspect	**Midterm Strategy (OIH/XES)**	**Final Strategy (Cross-Asset Pairs)**
Data Frequency	Minute-level	Daily-level
Asset Pairs	Single pair: OIH/XES	Four pairs: EEM/XLP, VWO/XLP, VWO/VDC, OIH/XES
Signal Type	Raw spread Z-score	Volatility-adjusted normalized spread
Signal Threshold	Static (1.5/2)	Static (0.005), momentum-filtered
Risk Controls	None	Stop-loss, take-profit, quick-stop, max holding
Trade Frequency	Very high	Low (avg 0.17% daily turnover)
Execution Cost	High ($5k+ in a week)	Low ($172 during live deployment)
Live Result (1 week)	-$57,000 (≈ -5.7%)	-$6,900 (≈ -0.7%)
Backtest Behavior	Unstable equity, noisy trades	Stable growth, tight drawdown
Portfolio Behavior	Unhedged exposure	Market-neutral with leverage control

Appendix A Strategy Implementation Code

```
from AlgorithmImports import *
import numpy as np
from datetime import timedelta

class CrossAssetRelativeMomentum(QCAlgorithm):
    """
    Cross-Asset Relative Momentum Strategy:
    Dynamically rotates long-short positions between Emerging Market Equities and
    Consumer Staples based on normalized relative performance,
    targeting short-term statistical divergence with controlled exposure via time-based
    and performance-based exits.
    """
    def Initialize(self):
        self.SetCash(1_000_000)
        self.SetStartDate(2022, 1, 1)
        self.SetBrokerageModel(BrokerageName.InteractiveBrokersBrokerage, AccountType.
    Margin)

        self.asset_pairs = [
            (self.AddEquity("OIH", Resolution.Daily).Symbol, self.AddEquity("XES",
    Resolution.Daily).Symbol),
            (self.AddEquity("EEM", Resolution.Daily).Symbol, self.AddEquity("XLP",
    Resolution.Daily).Symbol),
            (self.AddEquity("VWO", Resolution.Daily).Symbol, self.AddEquity("XLP",
    Resolution.Daily).Symbol),
            (self.AddEquity("VWO", Resolution.Daily).Symbol, self.AddEquity("VDC",
    Resolution.Daily).Symbol),
        ]

        self.lookback_window = 10
        self.signal_threshold = 0.005
        self.momentum_filter = 0.01
        self.position_leverage = 0.15
        self.holding_period = 5
        self.take_profit_threshold = 0.03
        self.stop_loss_threshold = -0.02
        self.quick_stoploss_threshold = -0.03

        self.price_history = {}
        self.in_position = {}
        self.entry_timestamp = {}
        self.entry_prices = {}

        self.total_trades = 0
        self.winning_trades = 0
        self.losing_trades = 0
        self.cumulative_pnl = 0.0
        self.open_trade_details = {}

        for pair in self.asset_pairs:
            self.price_history[pair] = []
```

```
            self.in_position[pair] = False
            self.entry_timestamp[pair] = None
            self.entry_prices[pair] = (None, None)
            self.open_trade_details[pair] = {}

        self.Schedule.On(self.DateRules.EveryDay(), self.TimeRules.AfterMarketOpen(self.
asset_pairs[0][0], 1), self.ExecuteTradingLogic)

    def safe_close(self, data, symbol):
        if not data.ContainsKey(symbol):
            return None
        if data[symbol] is None:
            return None
        return data[symbol].Close

    def OnData(self, data):
        for pair in self.asset_pairs:
            price1 = self.safe_close(data, pair[0])
            price2 = self.safe_close(data, pair[1])
            if price1 is None or price2 is None:
                continue
            self.price_history[pair].append((self.Time, price1, price2))
            if len(self.price_history[pair]) > self.lookback_window + 1:
                self.price_history[pair].pop(0)

    def ExecuteTradingLogic(self):
        for pair in self.asset_pairs:
            if len(self.price_history[pair]) < self.lookback_window + 1:
                continue

            times, prices1, prices2 = zip(*self.price_history[pair])
            ret1 = (prices1[-1] - prices1[-self.lookback_window]) / prices1[-self.
lookback_window]
            ret2 = (prices2[-1] - prices2[-self.lookback_window]) / prices2[-self.
lookback_window]
            spread_return = ret1 - ret2

            vol1 = np.std(np.diff(prices1[-self.lookback_window:]) / prices1[-self.
lookback_window:-1])
            vol2 = np.std(np.diff(prices2[-self.lookback_window:]) / prices2[-self.
lookback_window:-1])
            combined_vol = vol1 + vol2
            if combined_vol == 0:
                continue

            normalized_spread = spread_return / combined_vol
            if (abs(ret1) < self.momentum_filter) and (abs(ret2) < self.momentum_filter):
                continue

            if self.Portfolio.MarginRemaining < 2 * self.position_leverage * self.
Portfolio.TotalPortfolioValue:
                self.Debug(f"[{self.Time}] Insufficient margin for {pair}")
                continue
```

```
        if not self.in_position[pair]:
            if normalized_spread > self.signal_threshold:
                self.SetHoldings(pair[0], -self.position_leverage)
                self.SetHoldings(pair[1], self.position_leverage)
                self.in_position[pair] = True
                self.entry_timestamp[pair] = self.Time
                self.entry_prices[pair] = (prices1[-1], prices2[-1])
                self.open_trade_details[pair] = {'open_time': self.Time, 'price1':
prices1[-1], 'price2': prices2[-1]}
                self.Debug(f"[{self.Time}] Entered short {pair[0]} / long {pair[1]}")
            elif normalized_spread < -self.signal_threshold:
                self.SetHoldings(pair[0], self.position_leverage)
                self.SetHoldings(pair[1], -self.position_leverage)
                self.in_position[pair] = True
                self.entry_timestamp[pair] = self.Time
                self.entry_prices[pair] = (prices1[-1], prices2[-1])
                self.open_trade_details[pair] = {'open_time': self.Time, 'price1':
prices1[-1], 'price2': prices2[-1]}
                self.Debug(f"[{self.Time}] Entered long {pair[0]} / short {pair[1]}")

        else:
            holding_days = (self.Time - self.entry_timestamp[pair]).days
            current_price1 = prices1[-1]
            current_price2 = prices2[-1]
            entry_price1, entry_price2 = self.entry_prices[pair]

            if entry_price1 is None or entry_price2 is None:
                continue

            pnl_leg1 = (current_price1 - entry_price1) / entry_price1
            pnl_leg2 = (current_price2 - entry_price2) / entry_price2
            total_pnl = pnl_leg1 + pnl_leg2

            max_holding_days = self.holding_period

            if total_pnl < self.quick_stoploss_threshold:
                self.Debug(f"[{self.Time}] Quick stoploss triggered for {pair} at
total pnl {total_pnl:.4f}")
                self.close_position(pair, total_pnl, pnl_leg1, pnl_leg2)
                continue

            if (pnl_leg1 > self.take_profit_threshold) or (pnl_leg1 < self.
stop_loss_threshold) or \
               (pnl_leg2 > self.take_profit_threshold) or (pnl_leg2 < self.
stop_loss_threshold) or \
               (holding_days >= max_holding_days):
                self.close_position(pair, total_pnl, pnl_leg1, pnl_leg2)

def close_position(self, pair, total_pnl, pnl_leg1=0, pnl_leg2=0):
    self.Liquidate(pair[0])
    self.Liquidate(pair[1])
    self.in_position[pair] = False
    self.entry_timestamp[pair] = None
    self.entry_prices[pair] = (None, None)
```

```
        self.total_trades += 1
        if total_pnl > 0:
            self.winning_trades += 1
        else:
            self.losing_trades += 1
        self.cumulative_pnl += total_pnl

        win_rate = (self.winning_trades / self.total_trades) * 100 if self.total_trades >
 0 else 0

        self.Debug(f"Exited {pair} | Total Trades: {self.total_trades} | Wins: {self.
winning_trades} | Losses: {self.losing_trades} | Win Rate: {win_rate:.2f}% |
Cumulative PnL: {self.cumulative_pnl:.4f}")
        self.Debug(f"  Trade details: pnl_leg1 = {pnl_leg1:.4f}, pnl_leg2 = {pnl_leg2:.4f
}, open_time = {self.open_trade_details[pair].get('open_time')}")

    def record_trade_details(self, pair, pnl_leg1, pnl_leg2):
        pass
```